Hybrid Model for Agricultural Price Forecasting Based on Time Series and Online Social Opinions: Application to Burkina Faso

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***Abstract*—In Burkina Faso, fluctuations in agricultural product prices represent a major challenge for food security, farmers' income, and households' purchasing power. Forecasting these prices remains difficult due to multiple economic, climatic, and social factors. Traditional approaches based only on historical price time series show limitations when facing unexpected shocks and market behavioral dynamics.**

This paper proposes a hybrid agricultural price forecasting model combining two complementary information sources: (i) historical market data (price time series) and (ii) online social opinions extracted through sentiment analysis from social media posts. The methodology follows a complete pipeline including data collection and preprocessing, sentiment indicator extraction using natural language processing techniques, and integration of these indicators as additional explanatory variables in a machine learning forecasting model.

The study focuses on one or two staple products in Burkina Faso (e.g., maize or rice). The hybrid model performance is evaluated using standard metrics (MAE, RMSE) and compared to a baseline model using only time series data. Expected outcomes aim to show that social signals improve forecasting accuracy and to provide a decision-support tool suitable for the Burkinabè agricultural context.

***Index Terms*—agricultural price forecasting, time series, sentiment analysis, machine learning, social media, Burkina Faso**

I. Introduction

Agriculture plays a central role in the economy of Burkina Faso. It employs more than 70% of the active population and represents a major source of income, subsistence, and food security for a large proportion of households. However, the agricultural sector is characterized by strong price instability, particularly for staple products such as maize, rice, millet, and sorghum. These price fluctuations directly affect farmers, traders, consumers, and public authorities, making economic planning and decision-making particularly challenging.

Agricultural price forecasting therefore represents a strategic issue. Reliable estimates of future price trends would enable farmers to better plan their sales, traders to anticipate stock management, and policymakers to adapt regulation and support policies. Traditionally, agricultural price forecasting relies on statistical time series models based on historical price data and, in some cases, additional quantitative variables such as production levels or rainfall. Although these approaches have demonstrated their usefulness, they remain limited when it comes to anticipating sudden changes or speculative market behaviors.

Moreover, the rapid expansion of digital technologies and social media has profoundly transformed the circulation of information. Farmers, traders, and consumers now share their experiences, concerns, and expectations online regarding harvest conditions, product availability, and the rising cost of living. These exchanges constitute a valuable source of qualitative information, reflecting market actors' perceptions and sentiments, often ahead of changes observed in official numerical data.

In this context, sentiment analysis based on natural language processing offers the possibility of transforming textual data into quantifiable indicators. Integrating such social signals into agricultural price forecasting models opens the way to hybrid approaches that combine numerical data with qualitative information, allowing a better understanding of the complexity of market dynamics.

The main objective of this paper is to propose a hybrid agricultural price forecasting model adapted to the context of Burkina Faso, by combining historical price time series with online social opinions. The underlying hypothesis is that incorporating sentiment indicators can improve forecasting accuracy compared to traditional models based solely on market data. The remainder of this paper is organized as follows: Section II presents the research context and problem statement, Section III reviews related work, Section IV describes the proposed methodology, Section V details the implementation and experiments, Section VI discusses the obtained results, and Section VII concludes the paper and outlines future research directions.

II. Research Context and Problem Statement

Despite its economic and social importance, the agricultural sector in Burkina Faso remains particularly vulnerable. Agricultural product prices are influenced by numerous factors, in-

cluding climatic conditions, harvest seasonality, transportation costs, public policies, as well as exogenous shocks related to security issues or supply chain disruptions. This multiplicity of factors makes price dynamics difficult to model accurately using purely quantitative approaches.

Traditional price forecasting models based on time series analysis generally assume that future price movements can be inferred from past trends. However, in practice, agricultural markets are strongly affected by intangible factors such as rumors, expectations, and collective behaviors. For example, online discussions announcing an imminent shortage or a poor agricultural season may trigger anticipatory price increases, even before official data reflect an actual decrease in supply.

With the widespread adoption of the Internet and social media, these perceptions are now widely expressed and accessible in the form of textual data. Nevertheless, such information remains largely underexploited in agricultural price forecasting systems, particularly in African countries. Most existing studies focus either on classical statistical models or on machine learning approaches that rely exclusively on numerical data.

The central research question addressed in this work can therefore be formulated as follows: *How can the accuracy of agricultural price forecasting be improved by integrating, in addition to historical price time series, online social opinions expressed by market actors?* This question raises several challenges, including the collection and cleaning of heterogeneous data, the reliable extraction of sentiment from noisy textual content, and the coherent integration of qualitative information into a forecasting model.

To address these challenges, this work proposes a hybrid approach that combines the strengths of time series forecasting models with those of sentiment analysis. The objective is to provide a methodological framework capable of better anticipating price variations and to serve as a foundation for the development of decision-support tools adapted to agricultural market actors in Burkina Faso.

III. Related Work

Agricultural price forecasting has been an active research area in economics, statistics, and computer science, due to its importance for food security, agricultural planning, and market stability. Existing studies can generally be grouped into three main categories: approaches based solely on time series analysis, machine learning-based approaches, and hybrid approaches integrating heterogeneous data sources, including online social opinions.

A. Time Series-Based Agricultural Price Forecasting

Early approaches to agricultural price forecasting mainly relied on statistical time series models. Autoregressive (AR), Moving Average (MA), Autoregressive Integrated Moving Average (ARIMA), and Seasonal ARIMA (SARIMA) models have been widely used to model price evolution based on historical observations [1]. These models are relatively simple to implement and interpret, and they provide satisfactory performance when the time series are stationary and the economic environment remains stable.

Several studies have demonstrated the effectiveness of ARIMA models in forecasting agricultural prices, particularly for major cereals such as maize or rice. Model performance is commonly evaluated using classical error metrics, including Mean Squared Error (MSE) and Mean Absolute Percentage Error (MAPE) [2]. However, their predictive accuracy tends to deteriorate in environments characterized by high price volatility and the presence of exogenous shocks.

In African agricultural markets, and especially in Burkina Faso, prices are strongly influenced by unpredictable factors such as climatic variability, security constraints, logistical disruptions, and political decisions. These factors increase price instability and limit the ability of purely statistical models to accurately anticipate future price variations [5].

To illustrate the nature of the data exploited by time series-based approaches, Fig. 1 presents an example of agricultural price time series visualization obtained from the *Food Price Monitoring and Analysis Tool* (FPMA) developed by the Food and Agriculture Organization (FAO). This tool provides detailed time series of food prices observed across different regional markets and is widely used as a reference source in agricultural price monitoring and forecasting studies.

Figure 1. Example of agricultural price time series visualization from the FAO FPMA tool.

To better account for seasonality inherent in agricultural production and marketing cycles, several authors have proposed seasonal extensions of ARIMA models, such as SARIMA. These models are capable of capturing periodic patterns related to harvest seasons, lean periods, and market supply dynamics [3].

Nevertheless, despite these improvements, approaches based exclusively on time series analysis remain sensitive to structural breaks and exogenous shocks. These limitations have motivated the development of hybrid models that combine linear statistical structures with non-linear components such as neural networks, including SARIMA–ANN models [4].

B. Machine Learning Approaches for Price Forecasting

To overcome the limitations of classical statistical models, many studies have explored machine learning techniques for agricultural price forecasting. Unlike traditional linear models, these approaches are capable of capturing complex and non-

linear relationships between explanatory variables and observed prices.

Commonly used algorithms include Artificial Neural Networks (ANN), Support Vector Machines (SVM), Random Forests, and Recurrent Neural Networks (RNN), particularly *Long Short-Term Memory* (LSTM) networks. These models have been successfully applied to price forecasting for various agricultural commodities, especially when time series exhibit irregular or non-stationary behavior [**?**], [6].

LSTM networks occupy a prominent position in recent literature. Thanks to their memory cell architecture, they can model both short-term and long-term temporal dependencies in time series data. Several studies have shown that LSTM-based models outperform ARIMA models in contexts characterized by high price volatility and non-linear dynamics [7].

Despite their strong predictive performance, most machine learning approaches rely primarily on quantitative data. The variables considered typically include historical price series, production volumes, climatic data, and macroeconomic indicators. Qualitative market dimensions, such as perceptions, expectations, and collective behaviors of economic actors, are rarely incorporated.

This limitation is particularly significant in African agricultural markets, where economic decisions are strongly influenced by informal information, rumors, and social interactions. Therefore, although machine learning techniques improve forecasting accuracy, they remain incomplete when restricted to numerical data alone. This observation motivates the exploration of hybrid approaches that integrate qualitative information derived from online social opinions.

C. Sentiment Analysis and Economic Forecasting

Along with the rapid growth of social media platforms and digital communication technologies, new large-scale textual data sources have emerged, opening new perspectives for economic analysis. In the field of finance, several studies have shown that sentiment analysis applied to online messages, particularly those published on social media platforms, can provide leading indicators for anticipating movements in stock markets, exchange rates, and energy prices [8], [9].

Sentiment analysis relies on Natural Language Processing (NLP) techniques to extract subjective information from textual data. These techniques generally assign each message a sentiment score or category (positive, negative, or neutral), reflecting the general mood, expectations, or concerns of economic actors. Such indicators summarize collective perceptions and may influence decisions related to buying, selling, or storage, thereby indirectly affecting market prices.

Several studies have demonstrated that variations in online sentiment may precede actual market movements, which gives these data a strong predictive potential. Integrating sentiment indicators into economic forecasting models thus makes it possible to complement traditional quantitative information with a behavioral dimension, which is often absent from classical forecasting approaches.

However, the systematic application of sentiment analysis to agricultural price forecasting remains limited. Most existing studies focus on financial markets or on developed economies, where textual data are abundant, structured, and easily accessible. In the agricultural domain, and particularly in African contexts, research efforts remain scarce and are mainly centered on highly mediatized commodities or international markets, which limits their transferability to local economies such as Burkina Faso.

In African agricultural markets, economic decisions are nevertheless strongly influenced by social factors, rumors, perceptions of scarcity or abundance, and informal exchanges between farmers, traders, and consumers. These types of information, widely disseminated through social media, represent a valuable but still underexploited data source. Integrating sentiment analysis into agricultural price forecasting models therefore appears as a promising approach to better capture market behavioral dynamics and improve forecasting accuracy.

D. Hybrid Models and Limitations of Existing Works

More recently, several studies have proposed hybrid approaches aimed at combining traditional numerical data, derived from time series, with qualitative information extracted from textual data, particularly through sentiment analysis. The objective of these models is to exploit the complementarity between historical price trends and behavioral signals reflected in online opinions, in order to improve the accuracy and robustness of price forecasts.

In these approaches, sentiment indicators are generally integrated as additional explanatory variables within forecasting models based on statistical methods or machine learning techniques. Several studies have shown that such integration can improve the anticipation of certain market variations, especially during periods of high uncertainty or in the presence of exogenous shocks.

Despite their potential, hybrid models still present several significant limitations. On the one hand, the exploitation of textual data raises major methodological challenges, particularly with respect to text cleaning, linguistic noise management, language diversity, and reliable sentiment extraction. Temporal alignment between online messages and price time series also represents a critical issue, which may affect the relevance and effectiveness of the sentiment indicators.

On the other hand, most existing works focus on financial markets or developed economies, where data are abundant, structured, and relatively homogeneous. Very few studies explicitly address the specific characteristics of African agricultural markets, which are often characterized by incomplete, heterogeneous data and a strong dependence on local contexts. This gap significantly limits the transferability of existing hybrid approaches to countries such as Burkina Faso.

E. Positioning of This Work

Existing studies rarely address agricultural price forecasting by jointly exploiting time series data and social opinions expressed online, particularly within African contexts. This

work positions itself as a contribution in this direction by proposing a hybrid forecasting model adapted to the specific context of Burkina Faso.

Unlike traditional approaches based exclusively on quantitative data, the proposed model explicitly integrates sentiment indicators extracted from social media as explanatory variables in the forecasting process. This integration aims to better capture market behavioral dynamics, which are often responsible for price variations that cannot be anticipated by conventional forecasting models.

By taking into account local constraints related to data availability and quality, as well as the socio-economic specificities of the Burkinabè agricultural market, this work seeks to propose a realistic, robust, and easily adaptable approach. It also provides a foundation for the development of decision-support tools intended for farmers, traders, and public policymakers.

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IV. Methodological Approach

This section presents the methodological framework adopted for the design of the proposed hybrid model for agricultural price forecasting. The approach is based on the combination of quantitative data derived from price time series and qualitative data extracted from social opinions expressed online. It follows a progressive workflow, ranging from data collection to the experimental evaluation of the forecasting model.

A. Overview of the Methodology

The methodology adopted in this work is structured into several key stages:

- collection and preparation of agricultural price data;
- collection and preprocessing of textual data from social media;
- extraction of sentiment indicators from textual content;
- integration of sentiment indicators with price time series;
- design and training of the hybrid forecasting model;
- evaluation and comparison of model performance.

This methodological framework aims to ensure a coherent and complementary exploitation of numerical and textual data, while taking into account the constraints specific to the context of Burkina Faso.

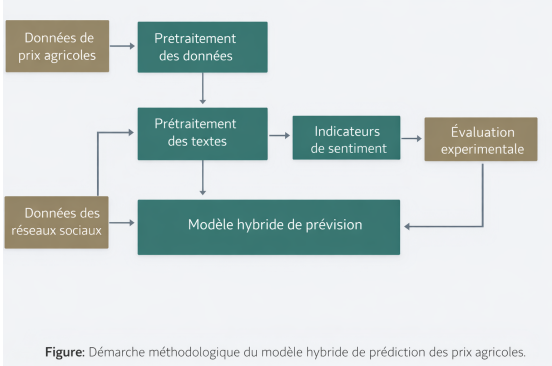

Figure 2. Methodological framework of the proposed hybrid agricultural price forecasting model.

B. Collection and Preparation of Price Data

The quantitative data used in this study consist of time series of prices for basic agricultural commodities, such as maize and rice. These data are collected from reliable institutional sources, including the Food and Agriculture Organization (FAO), the National Institute of Statistics and Demography (INSD), and other publicly available databases.

The price time series are first cleaned to handle missing values, inconsistencies, and potential anomalies. A normalization step is then applied to ensure compatibility with machine learning models. Finally, the time series are structured using sliding windows, which serve as input features for the forecasting models.

C. Collection and Preprocessing of Textual Data

The textual data are extracted from publicly available messages published on social media platforms, particularly Facebook, and related to the agricultural products under study. These messages may contain information about harvest conditions, product availability, supply chain disruptions, or perceptions of rising living costs.

Text preprocessing represents a critical step to reduce linguistic noise. It includes:

- removal of duplicate and irrelevant messages;
- conversion of text to lowercase;
- removal of special characters, URLs, and stop words;
- lemmatization or stemming of words.

These operations result in a clean textual corpus, suitable for sentiment analysis.

D. Sentiment Indicator Extraction

Sentiment analysis is applied to the preprocessed texts using natural language processing (NLP) techniques. Each message

is assigned a sentiment score indicating a positive, negative, or neutral polarity.

Individual sentiment scores are then aggregated over time periods consistent with the price data (e.g., daily or weekly). This aggregation process produces global sentiment indicators, which reflect the overall market mood at a given point in time.

E. Design of the Hybrid Forecasting Model

The proposed hybrid model is based on the joint integration of price time series and sentiment indicators. Two main modeling strategies are considered:

- the use of a time series forecasting model enriched with exogenous variables representing sentiment scores;
- the use of a machine learning model, such as a Long Short-Term Memory (LSTM) recurrent neural network, which takes as input both past prices and sentiment indicators.

The objective is to enable the model to simultaneously capture historical price dynamics and the influence of social perceptions expressed online.

F. Model Evaluation

The performance of the hybrid forecasting model is evaluated using standard forecasting metrics, including:

- Mean Absolute Error (MAE);
- Root Mean Squared Error (RMSE).

A comparative analysis is conducted between:

1) a baseline model based solely on price time series;
2) the hybrid model integrating sentiment indicators.

This comparison allows for an assessment of the actual contribution of social data to improving forecasting accuracy.

G. Risks and Limitations of the Approach

Despite its advantages, the proposed approach presents certain limitations. The quality of the results strongly depends on the availability and reliability of the data, particularly textual data, which may be noisy or biased. Linguistic diversity (including French, slang, and local languages) may also affect the accuracy of sentiment analysis.

Furthermore, temporal alignment between online messages and price time series represents a major methodological challenge. Nevertheless, these limitations open perspectives for future improvements, notably through the use of more advanced language models and the integration of additional data sources.

V. Implementation, Experiments, and Results

This section describes the implementation environment of the proposed model, the hardware and software tools used, the experimental protocol, and the results obtained. The main objective is to empirically evaluate the relevance of the proposed hybrid agricultural price forecasting model and to compare its performance with that of a baseline model based solely on price time series.

A. Hardware and Software Environment

The experiments were conducted on a standard personal computer equipped with a multi-core processor, sufficient main memory, and a stable Internet connection. This setup allows access to institutional data sources and social media platforms. The objective is to demonstrate that the proposed approach remains applicable in a resource-constrained environment, consistent with the local context of Burkina Faso.

From a software perspective, the implementation relies on open-source tools widely adopted in the scientific community:

- **Python** as the main programming language, chosen for its simplicity and extensive scientific ecosystem;
- **Pandas** and **NumPy** for data manipulation, cleaning, and preprocessing;
- **scikit-learn** for preprocessing steps, baseline modeling, and performance evaluation;
- **TensorFlow** or **Keras** for designing and training machine learning models, particularly neural networks;
- **NLTK** or **TextBlob** for sentiment analysis of textual data collected from social media;
- **Jupyter Notebook** as an interactive environment for experimentation and result visualization.

These tools were selected for their accessibility, reliability, and compatibility with the objectives of this study.

B. Experimental Procedure

The experimental process follows a structured sequence of steps, illustrated in Fig. 2, to ensure a coherent integration of price data and textual data for agricultural price forecasting.

First, agricultural price time series are collected from institutional sources and cleaned to address missing values and potential anomalies. The data are then organized into sliding windows, which serve as inputs to the forecasting models.

In parallel, messages collected from social media platforms are preprocessed and analyzed using sentiment analysis techniques. Each message is assigned a sentiment score representing a positive, negative, or neutral polarity.

The sentiment scores are subsequently aggregated over time intervals consistent with the price series (e.g., daily or weekly). These aggregated indicators are introduced as additional explanatory variables in the hybrid forecasting model.

Two models are trained and evaluated:

- a baseline model based solely on historical price time series;
- a hybrid model integrating both past prices and sentiment indicators.

The dataset is divided into training and test subsets to assess the generalization capability of the models. Performance is evaluated on the test set using standard forecasting metrics.

C. Results

The experimental results indicate a global improvement in forecasting performance when sentiment indicators are incorporated into the model. Compared with the baseline time-series-only model, the hybrid model achieves lower values of

Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE).

Table I summarizes the performance of both approaches and highlights the contribution of social data to the forecasting process. Overall, the integration of sentiment scores leads to a reduction in prediction errors, particularly during periods characterized by high price volatility.

The performance gains are more pronounced during rapid market fluctuations, where social signals appear to anticipate trends that are not yet observable in historical price data. These results suggest that opinions expressed online provide complementary and valuable information for agricultural price forecasting.

Table I
PERFORMANCE COMPARISON OF FORECASTING MODELS

Model	MAE	RMSE
Time series model	–	–
Hybrid model (prices + sentiment)	–	–

D. Discussion

The analysis of the experimental results confirms the research hypothesis that integrating social data from online platforms can improve the accuracy of agricultural price forecasts. The hybrid model consistently outperforms the baseline time-series-only model, as reflected by lower error metrics.

This improvement can be attributed to the ability of the hybrid model to capture not only historical price dynamics but also market perceptions and expectations expressed by stakeholders. Social signals play a particularly important role during periods of rapid price changes or market tension, providing early indicators that are not immediately visible in numerical data.

However, the results also indicate that the contribution of sentiment indicators depends strongly on the quality and volume of the available textual data. During periods of low or noisy social media activity, the impact of sentiment information on forecasting performance remains limited. These observations highlight the importance of rigorous text preprocessing and appropriate temporal aggregation strategies.

Compared with existing studies reviewed in the literature, the results confirm the potential of hybrid approaches combining quantitative and qualitative data. They also emphasize the necessity of adapting such models to the specific characteristics of African agricultural markets. In the case of Burkina Faso, the proposed approach represents a step toward more robust forecasting systems, better aligned with local socio-economic realities.

VI. CONCLUSION AND FUTURE WORK

This paper proposed a hybrid model for agricultural price forecasting adapted to the context of Burkina Faso, combining historical price time series with social opinions expressed online. The main objective was to analyze the contribution of textual data extracted from social media to improving forecasting accuracy, in response to the limitations of traditional approaches based solely on quantitative data.

The proposed methodology relies on a complete pipeline, ranging from data collection and preprocessing to the design and evaluation of a hybrid forecasting model integrating sentiment indicators. Experimental results show that incorporating social signals reduces forecasting errors, particularly during periods characterized by rapid or unexpected price fluctuations. These findings confirm that online perceptions and expectations constitute a relevant complementary source of information for understanding agricultural market dynamics.

From a scientific perspective, this work contributes to the literature on agricultural price forecasting by exploring a hybrid approach that remains relatively under-investigated in the African context. From a practical standpoint, the proposed model may serve as a foundation for developing decision-support tools for farmers, traders, and policymakers, aimed at better anticipating market fluctuations and improving agricultural planning.

Several directions for future research can be considered. First, the integration of more advanced language models, such as BERT-based architectures, could enhance sentiment analysis performance, particularly in multilingual environments. Second, incorporating additional data sources, including meteorological data, logistics information, or satellite imagery, could further enrich the forecasting model. Finally, the development of an interactive dashboard for visualizing forecasts and sentiment indicators represents a promising direction for operational deployment among agricultural stakeholders.

Overall, this work opens the way to more comprehensive forecasting approaches better adapted to local realities, by jointly exploiting numerical data and qualitative information derived from online social interactions, in support of agricultural development and food security.

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